

Client Project Management Requirements Handbook

[Student Version]



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[6th Edition, May. 2024]

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INTRODUCTION

This Project Management Requirements Handbook is a necessary tool for Wecloud faculty and students to establish how requirements will be collected, analyzed, documented, and managed throughout the lifecycle of client projects. Depending on the type of project there may be both project and product requirements. It is easy to unintentionally omit requirements, fail to document them, or leave requirements incomplete without a tool to properly manage them. Handbook can help you to better understand your responsibilities. It provides guidance to the faculty, students, project leads, and members who have administrative and technical oversight.

WeCloudData administration team issued this handbook in July 2020 and will update it on a yearly basis. Kindly contact us if you have any questions about expectations, scheduling, or any other concerns regarding the client projects.

Before reading:

Client project is one of the essential parts of the WeCloudData diploma program which helps students thrive on job applications. Unlike other companies, although students are not directly employed as full time workers, all the work that you did can be considered as your real working experience.

With that being said, individuals who are selected for client projects have to follow the policies and guidance where the deliverables and obligations should be committed (listed in below content). All your performances will be evaluated and shared with your pursuing organization or hiring agencies during background (performance) checking.

1. PROJECT SELECTION AND COMMITMENT

1.1 How are projects assigned to each student

Below is the general procedure:

1. **Beam Data** communicates all available projects to the project participants
 - A list of active projects
 - Project descriptions
 - Project expectations
 - Project commitment level
 - Project requirements suitability
2. Students will be assigned to 1-2 projects based on personal interests
 - Students will tell their preferred projects (industries) to the faculty member
 - **Note: putting in the preference doesn't mean a guaranteed reservation of spot**
3. Beam Data make the **1st project decision** based on the following criteria
 - Assessment test
 - Does the project align with the student's job search objective
 - E.g., we will not assign the student to an engineering-focused project if she/he wants a data science job
 - Can the student meet the time commitment minimum requirement
 - Does the student meet the technical requirements at the time of selection
 - If the project requires big data while the students haven't proven adequate skills then the student may not get selected. However, if there are not enough students interested in the project, then Beam Data will consider letting the student who has not acquired skills yet learn on the project
 - If the client requires filtering (resume, background) then we will communicate with the students and ask for a resume
 - If the client relies on Beam Data's selection then Beam Data will ask WeCloudData for students' academic performance
4. Beam Data makes the **2nd project decision** based on the following criteria

- Does the project align with the student's job search objective
 - E.g., we will not assign the student to an engineering-focused project if she/he wants a data science job
 - Can the student meet the time commitment minimum requirement
 - If a student is already involved in an ongoing client project, then the student will get less favored for the new project
 - eg. If the student has a parttime/fulltime job or other fairs that cannot guarantee your commitment, we'll also consider your available time spending on the project.
 - If a student is almost done with the 1st project, then we will put the student in the pool for consideration
 - If it's a really urgent request from a client then Beam Data may override the rules by selecting the student who has specific skills required for the project
 - E.g., if there's only one student knows Elasticsearch database then we may choose that student for help
 - How ready is the student for jobs?
 - If the student has not accomplished much on building the resume and portfolio, then we'd favor the student less because the faculty may want the student to focus on job search
 - Exceptions will be made if the student clearly communicated with us saying that she/he can delay the job search process in exchange for more project experience
 - How well did the student perform on the 1st project
 - 1st project performance assessment is a key factor for us to decide who to choose for a specific project (2nd project for the student)
5. All available slots for new projects will be announced to all the students. Final decision is made by the Principal (L1) and Project Manager (L2) and the BeamData faculty member. **We'll assign students to 2nd project based on the best overall fit and the project needs** (eg. 1st project performance, GPA, soft skills, time schedule)

If you want to wait for the “perfect” 2nd project it’s fine. But we encourage you to take the one that has immediate openings, which helps you fill the gap during your job search.

1.2 How many projects can a student work on?

Each student can work on 1-2 projects under the title of **Data Science Consultant**.

There will be special occasions that Beam Data may proactively bring some students to work on additional projects to meet the project deadlines. Who will be selected depends on a few things:

- Proficiency tests
- GPA
- Resume/Portfolio readiness (make sure your portfolio is ready for the screening)
- The student has special skills or background (e.g., data engineering, Elasticsearch, software engineering, etc.)

We discourage students to work on 2 projects at the same time. (*conditions applied)

If you have two projects that have time overlapping, it’s not necessary to tell the client that you are on multiple projects.

The 1st client project is **mandatory**, the 2nd is **optional**. You can start preparing your portfolio or focus on job searching at a certain point.

If you already get 2 interviews, we will stop assigning you projects as your qualifications are good enough for job searching.

1.3 How long can I stay on one project?

Each project will have its own minimum requirement of commitment. In general, each student has to commit at least **two months** for each project. Some projects require longer commitment time than others.

There are occasions where the project only requires a student to complete a specific task that gets completed in a few days. In that case, the student can discuss with the faculty to decide if the student can move on to another project sooner.

One can stay on projects **up to 6 months** after graduation so as to give opportunities to new students (Internal project is still opening and career support will be continue meanwhile)

1.4 Leaving the project

Project participants are expected to leave any project on good terms. Students will finish their commitment before wrapping up and starting the second project. If you leave a project in the first 3 weeks due to any reasons, you will not be considered as a team member on this project. We'll not provide any reference check related to this project.

- If you leave a project after a milestone has been completed, you will be eligible for performance evaluation. The evaluation will be used as part of your assessment at the job referral stage.
- Documentation of your part of the project must be completed before you leave the project

1.5 Removing from project

Each of our projects are the real ongoing industry projects and need your professionalism, meaning that you should treat it as your real job. **Your absence or inability to finish the assigned work should be communicated ahead, rather than a short notice.**

As we also have a long waiting list for many projects, students who

- Miss meetings with no reason for more than 2 times
- Unable to deliver the work for 2 weeks will be considered to be removed from the project.
- Receiving poor peer evaluations, but didn't get improved after communications

will be considered not capable enough to keep working on the project. (See 3.6 Attendance for details.)

In this case you'll be given 2 more weeks to finish whatever you wouldn't be able to finish, if you can finish it in time, you can still work on it till you finish.

2. ESCALATION PROCEDURE

Each project team will be managed by one of Beam Data's project managers and one student lead. Students who want to develop management skills would be considered as the project lead according to previous experiences.

When raising the issue or questions, project members will report to the lead first. If it cannot be solved internally, the project lead will escalate to the project manager.

2.1 Project Team Structure

- (L1) Principal
 - (L2) Project Manager 1
 - (L3) Project Team Lead
 - (L4) Team Member 1
 - (L4) Team Member 2
 - (L4) Team Member 3
 - (L2) Project Manager 2
 - (L3) Project Team Lead
 - (L4) Team Member 1
 - (L4) Team Member 2
 - (L4) Team Member 3

2.2 Escalation Flow

2.2.1 Internal Escalation

When encountering challenges during the project:

- Team member communicates with team lead and project manager directly
- Project lead escalate to the project manager if the team member approaches the project lead directly
- When necessary, the project manager will escalate the problem to the principal

2.2.2 External Escalation

When encountering challenges during the project:

- If it's data and process related issues team member can contact the engineers/scientists/managers on the client side directly
 - However, the team member needs to keep the team lead in the loop
 - Team lead needs to communicate with the project manager in review meetings
- If the challenge or question is data strategy related that will affect the workflow,
 - Team members need to communicate the problem with team lead and project manager first
 - The project manager needs to inform the principal when necessary

2.3 Election of the team leader

Team members can volunteer to become the team leader. In cases when there are multiple team members interested in being the team leader, the project manager and the principal will decide based on the following criteria

- Work experience and background
- GPA
- Communication skills
- Past project performance if applicable

3. PROJECT COMMUNICATIONS

Communication is highly important for the success of any project. Students are required to communicate in a professional manner.

3.1 Onboarding Training

- Self-introduction
 - [You'll work as Data Science \(or Data Engineer\) Consultant at Beamdata](#)
 - Can briefly mention your past experiences but keep data relevant works
 - Treat yourself as a mature consultant:

You are joining to work and help, rather than 'I'm here to learn' - Clients expect someone who can deliver things right away instead of waiting for training from the scratch.

- Time conflicts or disagreement on the task planning
 - It's understandable sometimes you cannot attend the client meeting. You'll escalate to project lead at the first time and then keep the project manager noticed.
 - If you get interviews (or you have a full time job or other conflicts), **there's no need to mention them in front of the client.** They are aware of our situation but it's not proper to use 'I have to finish my other work' or 'I am preparing for an interview' as the excuse that you will not deliver the assigned tasks.

3.2 Emails, slack messages

- Email communication
 - For all external communication (with the client), email is the preferred communication channel for any project communications
 - Who to include on the email cc list
 - Client
 - PM
 - Team lead
 - Understand the difference between CC vs BCC
 - Watch the tones of your communication, always be positive, and provide enough context in your communication.
 - **No skip level emails!** - Send emails before getting your project manager's approval
- **Slack communication**
 - Used for interactive technical discussions, and notifications
 - Even when the client is involved in the slack discussion, a formal email should be sent as a follow-up if applicable
- Zoom
 - General meetings (internal & external) will be held using zoom

- Team members will share screen to present outcomes
- The team lead can apply for session recording if it's needed.

3.3 Communicating technical issues

- Always communicate the context of your questions ([see 4.1](#))

3.4 Conflicts and complaints

Conflicts of interest and opinions are normal during a project. Here's the suggested way to resolve conflicts, express concerns, and make complaints.

3.4.1 Conflict of interest or opinions

- Always try to communicate openly in a professional manner. Communicate properly and express your opinions in a calm and professional manner with the counterparty that you have conflict or different opinion with
- If the problem is not resolved, try to communicate with team lead first and then escalate to the project manager if necessary

3.4.2 Complaints

- If you think there's unfairness in the process or some other team members are not contributing, let the project lead and PM know
- PM and the team lead need to handle the conflict/complaints in an unbiased manner
- **Backstabbing and badmouthing other team members is not our friend!**

3.4.2 Schedule conflicts

- If you have upcoming interviews that cause schedule conflict, you're responsible for letting the project team know beforehand whenever possible.
- In case of emergency, try your best to inform the team lead and project manager
- Proper handover is always required and the students need to try the very best to make sure that the tasks can be completed on time

3.5 Disagreements on approaches and methodologies

- Communicate with team lead and then escalate to PM

3.6 Attendance

Students have to be dedicated to stand-up and weekly meetings. **If one is absent for more than 2 meetings without any proper communication, we will consider your continued participation in the project.** (see 5.2 Internal meetings)

- The project lead is responsible for record meeting attendance
- Coordinate with the project lead or PM ahead of schedule if you can not attend or finish the work right away for any reason
- Beam Data won't provide the background checking for you in the project

4. TYPES OF QUESTIONS OR CONCERNS

4.1 Technical question (model, EDA)

Students are required to complete the tasks independently as a data scientist. When encountering syntax errors, technical challenges students are expected to do research and solve the problems on their own. **Project managers will not answer basic technical questions when you encounter them during the client project.**

We would like to give guidance and will supervise the general scope throughout the project. To save everyone's time, it's necessary to go through those when you describe the issue:

1. The problem definition (The problem details that block you)
2. What efforts that you've tried to solve
3. What are the corresponding results after trying

After explaining the above details we can better understand your situation and help you to overcome difficulties.

4.2 Time management

One has to inform both project lead and project managers once there are some time conflicts with team meetings.

Nevertheless, if clients want an urgent delivery which is hard to accomplish, it is students' right to let the project manager know about the situation. The project manager will communicate on behalf of all team members and coordinate with the client's needs.

4.3 Tasks arrangement

Project managers will assign tasks according to each member's skill sets. For example, A is focusing on model development while B is good at building visualizations. Please keep managers informed about your expectations and expertise throughout the projects.

5. DOCUMENTATION

In order to effectively manage a project, communication must be managed and controlled. Additionally, every effort must be taken to identify requirements thoroughly during project initiation and planning. However, there are often situations which require changes to a project or its requirements. In these situations it is important to utilize configuration management to consider proposed changes, establish a process to review and approve any proposed changes, and to implement and communicate these changes to the clients.

We are using **Notion** for project management.

5.1 Task assignments and tracking

Project Manager (L2) and Project Lead (L3) will be responsible for creating a weekly or bi-weekly Sprint Plan. Project Lead (L3) must make sure to create and assign tasks in the Notion kanban board to team members (L4) according to their expertise or capability discussed under your internal meeting. Then Team members (L4) could change your task status and log the time spent on your tasks accordingly.

To indicate your status online is mandatory. Your performance will be also evaluated by the pulled weekly time logging report.

5.2 Internal team meetings

[WeCloudData / BeamData Project Meeting Guidelines](#) describe all the meeting types and requirements.

5.2.1 Daily stand up meeting (internal)

A daily 10-minute meeting for the teams to communicate the progress. Less than 30 minutes. **Project Lead (L3) will do the attendance check in a public board.** Skipping more than 2 internal meetings without any reason, you'd be considered not qualified for continuing the project.

5.2.1 Sprint planning meeting

Weekly meetings for the team to discuss accomplished tasks and plan for the next sprint.

Each group has to log your notes to Notion during or after your meetings. Project Lead (L3) will create a meeting template under a certain section and Team members (L4) are responsible for creating your own part.

Everything should be documented. It is essential to list all the questions and decision items in the meeting notes and will be helpful for your client meetings.

Detailed guidance is provided in the file of [BeamData_Group_meeting_template.pdf](#)

5.2.3 Weekly update meeting with PM *

The Project Manager (L2) will have a weekly meeting with each Project Team Lead (L3) for project coordination and updates. Before the meeting, Project Team Lead (L3) is responsible for contributing to a weekly meeting report with the meeting agenda.

Meeting template is the same as the regular team meeting notes

[BeamData_Group_meeting_template.pdf](#)

5.3 Slides / Dashboards

Slides or dashboard templates will be provided in Notion. Slides contributors could upload your presentation files to the pre-assigned section for future reference.

Team Member (L4) is also responsible for providing their **individual project (milestone) documentation page** in Notion during the project and before wrapping up.

5.4 Milestone achievements *

Milestone is normally planned by clients. It varies between different projects. Project leads (L3) will help with Project Manager (L2) with the **sprint planning and tasks** assignments during the whole process.

* **Applied to project leads**

6. PERFORMANCE EVALUATION (GPA AND REFERRAL)

Students' performance in the project will be evaluated and the project performance score will become part of the GPA.

As a prerequisite, students will commit to the project tasks according to the project plan.

If you don't work hard on it due to any reasons, you won't get tasks assigned in the future, hence your true performance will also be reflected in your background checking*.

(*Background checking cases please see [Client Project Requirements Case Sharing-Supplementary document of Project Handbook.pptx](#))

Project performance will affect the following:

1. GPA
2. Selection of future client projects
3. Referral

The Project Performance Score is a weighted score of peer review, project manager review, and client review.

Weclouddata/Beamdata will definitely support and give you recommendations when the future employers contact us. However, we'll tell the **true story** about your performance during the background checking.

6.1 Peer review (30%)

Assessment score:

- Attendance
- How well does your peer communicate during the project
 - Active during the internal and external meeting
 - Problem solving skills
 - Willingness to help your peers
- Your commitment to the project
- Overall contribution
- **Documentation**

Peer comments:

- Eg. Improve communication skills / Should practise time management skills etc.

Your 1st project review will affect the decision your 2nd project selection

6.2 Manager review (40%)

- Team lead review
- PM review
- Instructors

6.3 Client Review (team level) (30%)

- We ask clients to review the entire project team when finished.

All the reviews are expected to be used as the reference in your background checking process.

7. CONFIDENTIALITY

7.1 Non-Disclosure Agreement

- All students need to sign the NDA form provided by the client or BeamData before starting the project.
- Things to avoid
 - Don't share client information
 - Don't disclose list of clients of beam data
 - Don't publish any client data
 - Don't use Tableau/PowerBI (Free Version) for client projects.

7.2 What information can be shared on portfolio/resume/linkedin

Always communicate with the PM or the client for the information that you want to share on your public profile.

- DOS:
 - Key achievements
 - Developed the customer churn model using customer data
 - Created the automation process for the product price scraping
 - etc.
 - Visualizations
 - Public data sources
 - Model performances
- **DON'TS**
 - New business plan
 - New product line release
 - Confidential internal information (customer emails, names...)
 - Ongoing marketing strategies (exact promotion on A product)
 - Business Strategies
 - Don't share client project details on the portfolio (github, blogs etc.)

Focus on your client project (let us know if you are doing projects sponsored by other organization)

7.3 What can be shared in interviews

Similar as 7.2.

Students can also practise on the mock interview during the Career Support Sessions (Week13 in general).

8. Graduation Defence

Project Milestone Presentation